



Combating Political Deepfakes with AI Detection and Human Moderation

Team 18: Luis Cantoran, Julia Lasiota, Jonathan Ocampo, Zareef Shafquat, Caeley Woo

CS152/COMM122/INTLPOL267: Trust and Safety

Department of Computer Science, Stanford University

Problem Description

The widespread availability of AI tools makes it fast, cheap, and easy to produce fake imagery of real political figures and circulate them on social media. With each release of new models capable of producing more hyperrealistic images, the problem is only accelerating. The attackers, ranging from malign individuals to government-linked networks, target both the political figures as well as the general public. Recent research found that AI images deployed in a political context can undermine trust in elected representatives, where 60% of viewers believed a political deepfake was real [1]. These concerns are timely given that social media is becoming increasingly relevant for shaping political discourse—over half of U.S. adults report that they sometimes get news from online platforms [2].

Policy Language

We want our platform to be a space where users can consume content without worrying about its veracity. Due to the rise in convincing AI-generated deepfake images of politicians, we are seeking to both respect our users’ ability to post what they would like while also prohibiting the posting of any AI-generated deepfakes of political figures.

Our platform is taking the following measures to enforce this policy:

- Only moderate images of politicians:** currently, we are focused primarily on combating the spread of political deepfake images. As a result, we are not policing AI images of your baby as a superhero. We use a ML service (Rekognition) trained to detect whether or not any faces in your image belong to a politician or not.
- Only moderate images with high confidence of being AI:** Real political content is welcome on our app! We leverage OpenAI Vision to produce a likelihood score of whether or not your image is AI. We will only send your image to a human moderator if the combined scores from 1) likelihood of containing a politician and 2) liklihood of being AI are high enough.
- Leave the final takedown decisions to a human:** We care about not restricting your ability to post authentic images. As such, only real, human moderators have the ability to take down a post altogether. Until a human moderator has made a decision, posts flagged as likely AI will be hidden from a wider audience but remain visible to your friends.

If you violate this policy, we reserve the right to remove your image, flag it as AI for all future decisions, and flag your account as being in violation of our AI policy. We also reserve the right suspend or ban any accounts found to be in violation of our AI policy more than three times, at the discretion of our moderators.

Evaluation

	Precision	Recall	F1	USD/1k	Latency Median/P99
Heuristic	0.471	0.160	0.239	1.00	0.927
Traditional ML	0.706	0.600	0.649	0.00	0.927
LLM-as-classifier	1.000	0.030	0.058	13.00	0.927
Hybrid: ML + LLM	0.769	0.500	0.606	3.25	0.927
Hybrid: heuristic + LLM	1	0.470	0.639	13.00	0.927

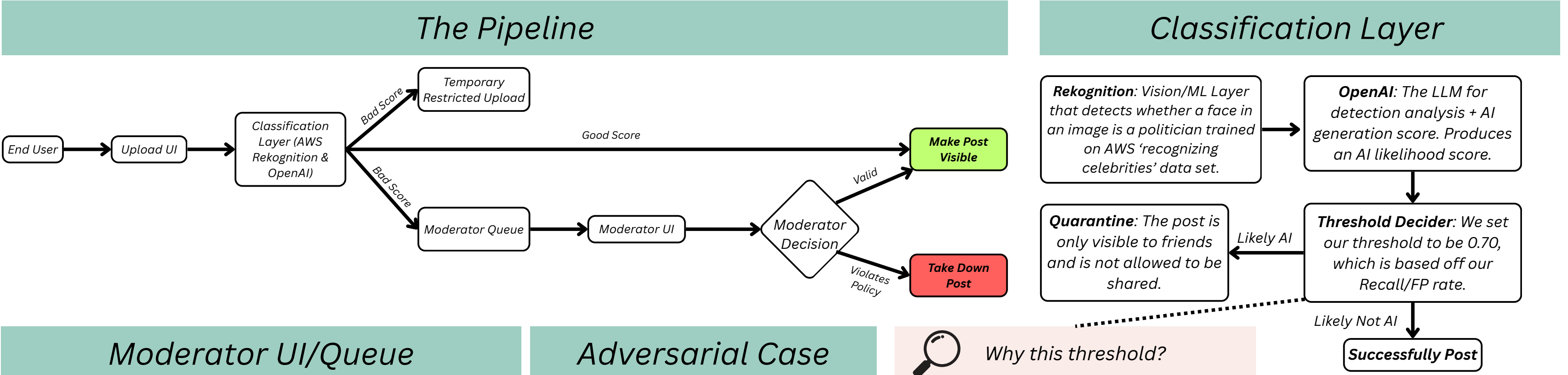
Confusion Matrix for Production

	Flag	Allow
Fake	47	53
Real	0	100

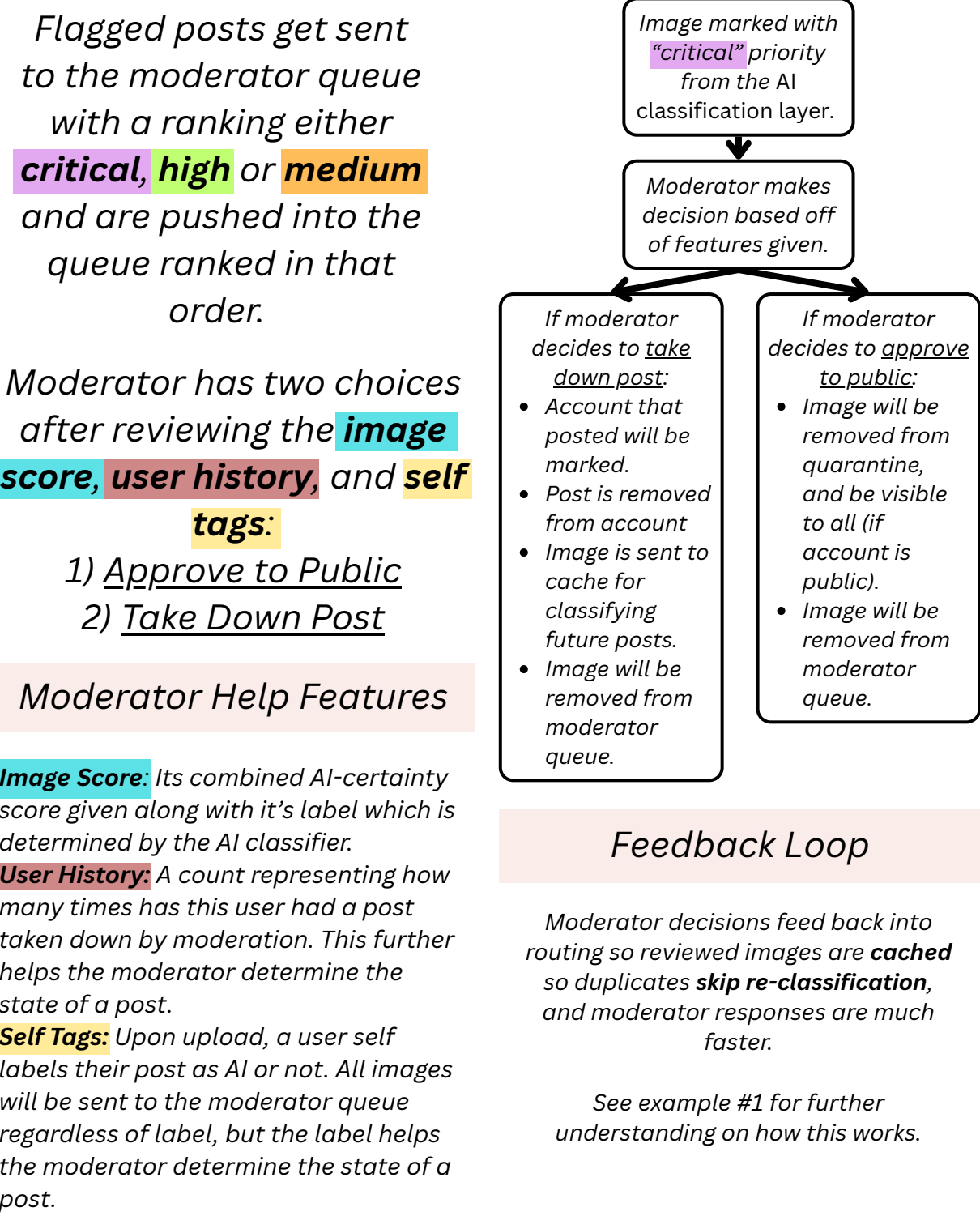
Key findings -

- Our hybrid of heuristic + LLM performed the best (relatively strong recall and perfect precision) and is what we used in production.
- While several approaches have perfect precision, recall is relatively low across the board, meaning that real images are hardly ever being flagged as false but fake images are somewhat likely to get through.

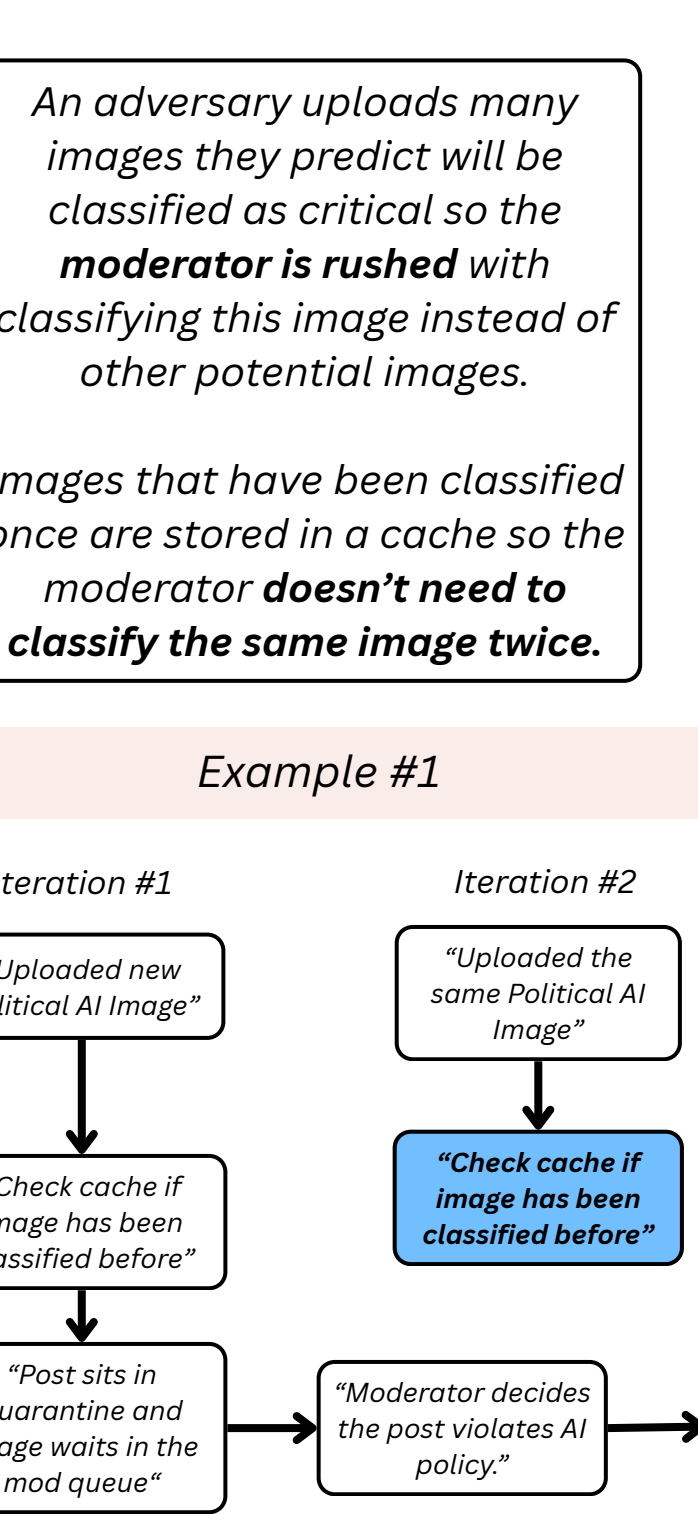
Technical Back-End



Moderator UI/Queue



Adversarial Case



Looking Forward

We intentionally kept our scope rather narrow for this project – namely, preventing the spread of AI-generated political disinformation images. With more time, we think that it would be important to detect not just images of political disinformation but also videos, particularly as deepfake videos become increasingly realistic. Similarly, while we found policial deepfakes to be an important starting point for deepfake detection (because of the real-life implications of people perceiving these deepfakes as real), eventually we would like to create a more general version of our classifier that can handle all kinds of deepfakes. A remaining concern is that also want to be cognizant of the costs of our solution at scale. As noted in the evaluation section, we found that running a traditional ML classifier did a poor job at detecting political deepfakes and so our current pipeline relies on calls to an LLM (OpenAI Vision) with every post upload that is classified as an image of a politician; this will be expensive with scale.